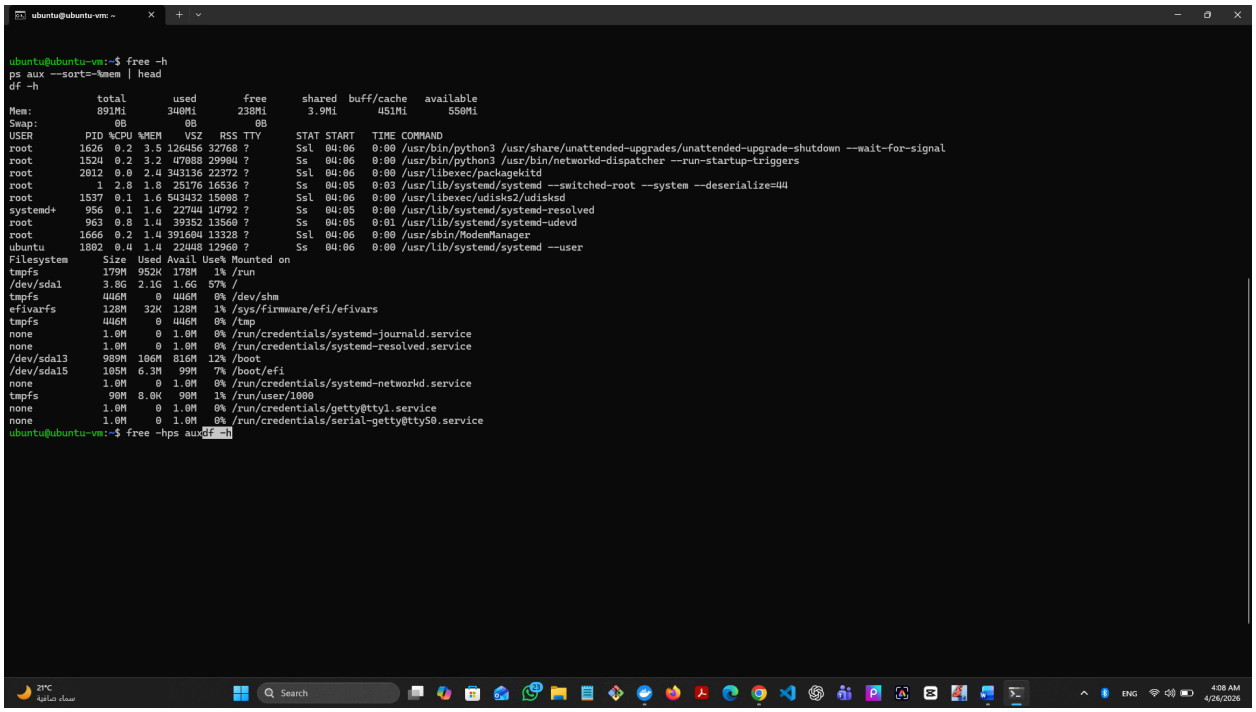
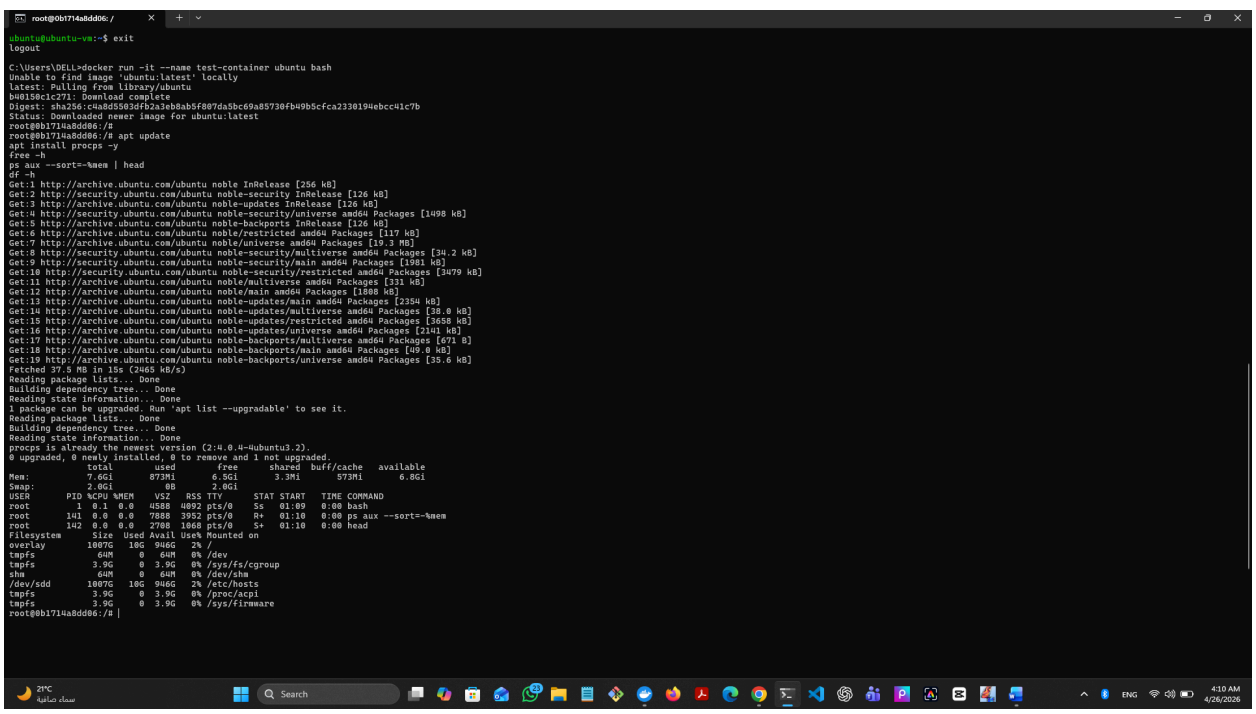
Lab1

Terminal Outputs:



```
ubuntu@ubuntu-vm: ~
ubuntu@ubuntu-vm:~$ free -h
ps aux --sort=-%mem | head
df -h
              total        used        free      shared  buff/cache   available
Mem:           891Mi       340Mi       238Mi       3.9Mi       451Mi       550Mi
Swap:             0B          0B          0B
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root        1626  0.2  3.5 126456 32768 ?        Ssl  04:06   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root        1524  0.2  3.2  47088 29904 ?        Ss   04:06   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
root        2012  0.0  2.4 343136 22372 ?        Ssl  04:06   0:00 /usr/libexec/packagekitd
root           1  2.8  1.8  25176 16536 ?        Ss   04:05   0:03 /usr/lib/systemd/systemd --switched-root --system --deserialize=44
root        1537  0.1  1.6 543432 15008 ?        Ssl  04:06   0:00 /usr/libexec/udisks2/udisksd
systemd+     956  0.1  1.6  22744 14792 ?        Ss   04:05   0:00 /usr/lib/systemd/systemd-resolved
root         963  0.8  1.4  39352 13560 ?        Ss   04:05   0:01 /usr/lib/systemd/systemd-udevd
root        1666  0.2  1.4 391604 13328 ?        Ssl  04:06   0:00 /usr/sbin/ModemManager
ubuntu      1802  0.4  1.4  22448 12960 ?        Ss   04:06   0:00 /usr/lib/systemd/systemd --user
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           179M  952K  178M   1% /run
/dev/sda1       3.8G  2.1G  1.6G  57% /
tmpfs           446M     0  446M   0% /dev/shm
efivarfs        128M   32K  128M   1% /sys/firmware/efi/efivars
tmpfs           446M     0  446M   0% /tmp
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
/dev/sda13      989M  106M  816M  12% /boot
/dev/sda15      105M  6.3M   99M   7% /boot/efi
none            1.0M     0  1.0M   0% /run/credentials/systemd-networkd.service
tmpfs            90M  8.0K   90M   1% /run/user/1000
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
none            1.0M     0  1.0M   0% /run/credentials/serial-getty@ttyS0.service
ubuntu@ubuntu-vm:~$ free -hps auxdf -h
```



```
root@0b1714a8dd06: /
ubuntu@ubuntu-vm:~$ exit
logout

C:\Users\DELL>docker run -it --name test-container ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
b40150c1c271: Download complete
Digest: sha256:c4a8d5503dfb2a3eb8ab5f807da5bc69a85730fb49b5cfca2330194ebcc41c7b
Status: Downloaded newer image for ubuntu:latest
root@0b1714a8dd06:/#
root@0b1714a8dd06:/# apt update
apt install procps -y
free -h
ps aux --sort=-%mem | head
df -h
Get:1 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1498 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [34.2 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1981 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [3479 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2354 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [38.0 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3658 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [2141 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [671 B]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [49.8 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [35.6 kB]
Fetched 37.5 MB in 15s (2465 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
1 package can be upgraded. Run 'apt list --upgradable' to see it.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
procps is already the newest version (2:4.0.4-4ubuntu3.2).
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
              total        used        free      shared  buff/cache   available
Mem:           7.6Gi       873Mi       6.5Gi       3.3Mi       573Mi       6.8Gi
Swap:          2.0Gi          0B       2.0Gi
USER       PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.1  0.0   4588  4092 pts/0    Ss   01:09   0:00 bash
root       141  0.0  0.0   7888  3952 pts/0    R+   01:10   0:00 ps aux --sort=-%mem
root       142  0.0  0.0   2708  1068 pts/0    S+   01:10   0:00 head
Filesystem      Size  Used Avail Use% Mounted on
overlay        1007G   10G  946G   2% /
tmpfs            64M     0   64M   0% /dev
tmpfs           3.9G     0  3.9G   0% /sys/fs/cgroup
shm              64M     0   64M   0% /dev/shm
/dev/sdd       1007G   10G  946G   2% /etc/hosts
tmpfs           3.9G     0  3.9G   0% /proc/acpi
tmpfs           3.9G     0  3.9G   0% /sys/firmware
root@0b1714a8dd06:/#
```

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\System32> Measure-Command { multipass launch --name test-vm2 }


Days              : 0
Hours             : 0
Minutes           : 1
Seconds           : 30
Milliseconds      : 677
Ticks             : 906777065
TotalDays         : 0.00104951049189815
TotalHours        : 0.0251882518055556
TotalMinutes      : 1.51129510833333
TotalSeconds      : 90.6777065
TotalMilliseconds : 90677.7065



PS C:\WINDOWS\System32>
PS C:\WINDOWS\System32> Measure-Command { docker run ubuntu echo hello }


Days              : 0
Hours             : 0
Minutes           : 0
Seconds           : 1
Milliseconds      : 343
Ticks             : 13436467
TotalDays         : 1.55514664351852E-05
TotalHours        : 0.000373235194444444
TotalMinutes      : 0.0223941116666667
TotalSeconds      : 1.3436467
TotalMilliseconds : 1343.6467


PS C:\WINDOWS\System32>
```
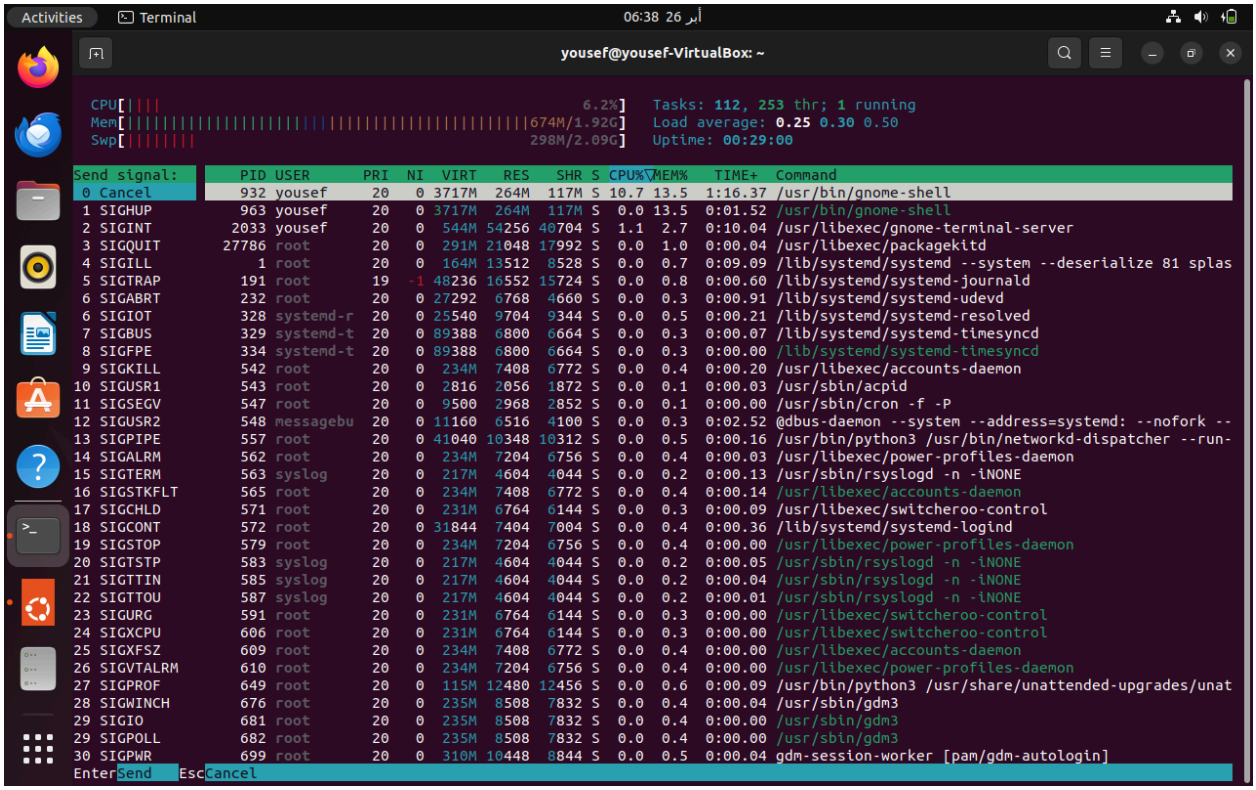
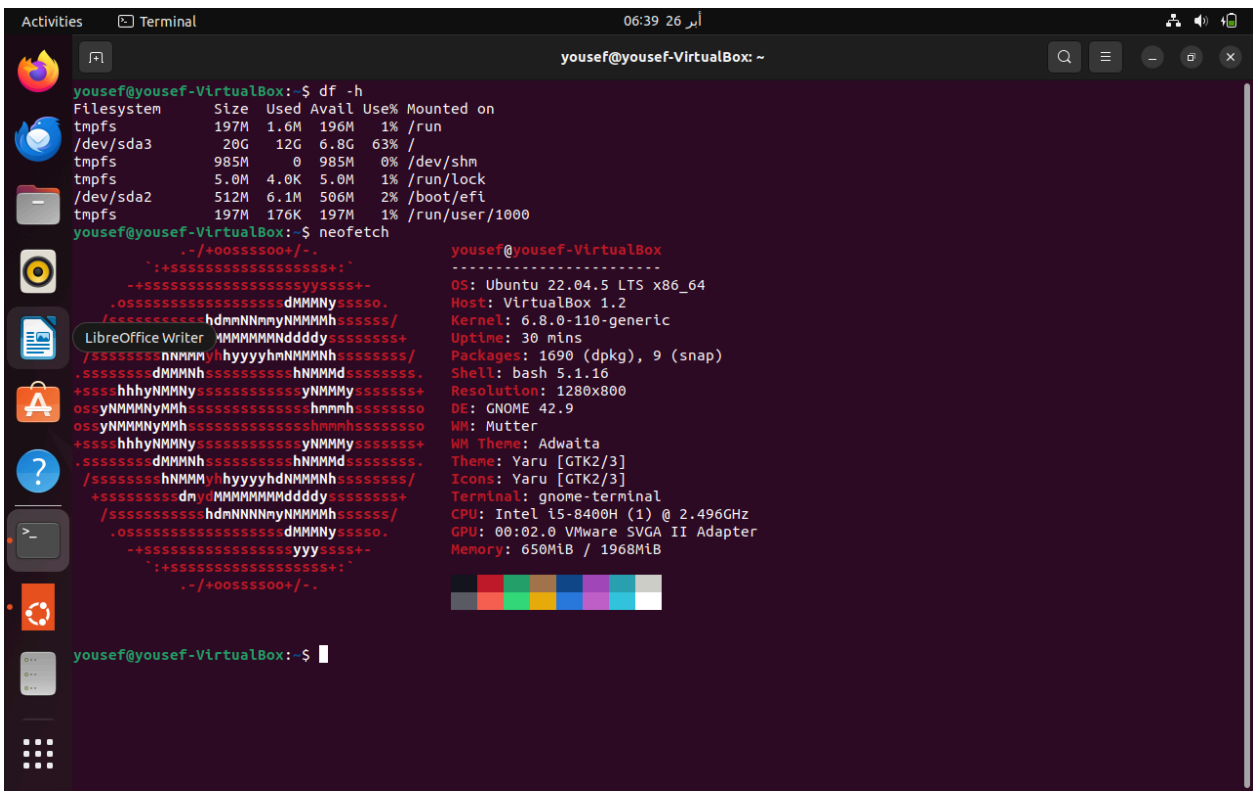

VM outputs:

```
Activities   Terminal   06:36 26 أبر
yousef@yousef-VirtualBox: ~
                                ase bmi1 bmi2 invpcid rdseed adx clflushopt arat md
                                _clear flush_l1d arch_capabilities
Virtualization features:
  Hypervisor vendor:            KVM
  Virtualization type:          full
Caches (sum of all):
  L1d:                          32 KiB (1 instance)
  L1i:                          32 KiB (1 instance)
  L2:                           256 KiB (1 instance)
  L3:                           8 MiB (1 instance)
NUMA:
  NUMA node(s):                 1
  NUMA node0 CPU(s):            0
Vulnerabilities:
  Gather data sampling:         Not affected
  Indirect target selection:    Mitigation; Aligned branch/return thunks
  Itlb multihit:                Not affected
  L1tf:                         Mitigation; PTE Inversion
  Mds:                          Mitigation; Clear CPU buffers; SMT Host state unkno
                                wn
  Meltdown:                     Mitigation; PTI
  Mmio stale data:              Mitigation; Clear CPU buffers; SMT Host state unkno
                                wn
  Reg file data sampling:       Not affected
  Retbleed:                     Vulnerable
  Spec rstack overflow:         Not affected
  Spec store bypass:            Vulnerable
  Spectre v1:                   Mitigation; usercopy/swapgs barriers and __user poi
                                nter sanitization
  Spectre v2:                   Mitigation; Retpolines; STIBP disabled; RSB filling
                                ; PBRSB-eIBRS Not affected; BHI Retpoline
  Srbds:                        Unknown: Dependent on hypervisor status
  Tsa:                          Not affected
  Tsx async abort:              Not affected
  Vmscape:                      Not affected
yousef@yousef-VirtualBox:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:          1.9Gi       671Mi       233Mi        17Mi       1.0Gi       1.1Gi
Swap:         2.1Gi       298Mi       1.8Gi
yousef@yousef-VirtualBox:~$
```


Activities
Terminal
06:38 26 أبر
yousef@yousef-VirtualBox: ~
CPU[|||| 6.2%]
Mem[||||||||||||||||||||||||||||||||||||||||||||||||||||674M/1.92G]
Swp[|||||||| 298M/2.09G]
Tasks: 112, 253 thr; 1 running
Load average: 0.25 0.30 0.50
Uptime: 00:29:00
Send signal:
0 Cancel
1 SIGHUP
2 SIGINT
3 SIGQUIT
4 SIGILL
5 SIGTRAP
6 SIGABRT
6 SIGIOT
7 SIGBUS
8 SIGFPE
9 SIGKILL
10 SIGUSR1
11 SIGSEGV
12 SIGUSR2
13 SIGPIPE
14 SIGALRM
15 SIGTERM
16 SIGSTKFLT
17 SIGCHLD
18 SIGCONT
19 SIGSTOP
20 SIGTSTP
21 SIGTTIN
22 SIGTTOU
23 SIGURG
24 SIGXCPU
25 SIGXFSZ
26 SIGVTALRM
27 SIGPROF
28 SIGWINCH
29 SIGIO
29 SIGPOLL
30 SIGPWR
PID USER PRI NI VIRT RES SHR S CPU%▽MEM% TIME+ Command
932 yousef 20 0 3717M 264M 117M S 10.7 13.5 1:16.37 /usr/bin/gnome-shell
963 yousef 20 0 3717M 264M 117M S 0.0 13.5 0:01.52 /usr/bin/gnome-shell
2033 yousef 20 0 544M 54256 40704 S 1.1 2.7 0:10.04 /usr/libexec/gnome-terminal-server
27786 root 20 0 291M 21048 17992 S 0.0 1.0 0:00.04 /usr/libexec/packagekitd
1 root 20 0 164M 13512 8528 S 0.0 0.7 0:09.09 /lib/systemd/systemd --system --deserialize 81 splas
191 root 19 -1 48236 16552 15724 S 0.0 0.8 0:00.60 /lib/systemd/systemd-journald
232 root 20 0 27292 6768 4660 S 0.0 0.3 0:00.91 /lib/systemd/systemd-udevd
328 systemd-r 20 0 25540 9704 9344 S 0.0 0.5 0:00.21 /lib/systemd/systemd-resolved
329 systemd-t 20 0 89388 6800 6664 S 0.0 0.3 0:00.07 /lib/systemd/systemd-timesyncd
334 systemd-t 20 0 89388 6800 6664 S 0.0 0.3 0:00.00 /lib/systemd/systemd-timesyncd
542 root 20 0 234M 7408 6772 S 0.0 0.4 0:00.20 /usr/libexec/accounts-daemon
543 root 20 0 2816 2056 1872 S 0.0 0.1 0:00.03 /usr/sbin/acpid
547 root 20 0 9500 2968 2852 S 0.0 0.1 0:00.00 /usr/sbin/cron -f -P
548 messagebu 20 0 11160 6516 4100 S 0.0 0.3 0:02.52 @dbus-daemon --system --address=systemd: --nofork --
557 root 20 0 41040 10348 10312 S 0.0 0.5 0:00.16 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-
562 root 20 0 234M 7204 6756 S 0.0 0.4 0:00.03 /usr/libexec/power-profiles-daemon
563 syslog 20 0 217M 4604 4044 S 0.0 0.2 0:00.13 /usr/sbin/rsyslogd -n -iNONE
565 root 20 0 234M 7408 6772 S 0.0 0.4 0:00.14 /usr/libexec/accounts-daemon
571 root 20 0 231M 6764 6144 S 0.0 0.3 0:00.09 /usr/libexec/switcheroo-control
572 root 20 0 31844 7404 7004 S 0.0 0.4 0:00.36 /lib/systemd/systemd-logind
579 root 20 0 234M 7204 6756 S 0.0 0.4 0:00.00 /usr/libexec/power-profiles-daemon
583 syslog 20 0 217M 4604 4044 S 0.0 0.2 0:00.05 /usr/sbin/rsyslogd -n -iNONE
585 syslog 20 0 217M 4604 4044 S 0.0 0.2 0:00.04 /usr/sbin/rsyslogd -n -iNONE
587 syslog 20 0 217M 4604 4044 S 0.0 0.2 0:00.01 /usr/sbin/rsyslogd -n -iNONE
591 root 20 0 231M 6764 6144 S 0.0 0.3 0:00.00 /usr/libexec/switcheroo-control
606 root 20 0 231M 6764 6144 S 0.0 0.3 0:00.00 /usr/libexec/switcheroo-control
609 root 20 0 234M 7408 6772 S 0.0 0.4 0:00.00 /usr/libexec/accounts-daemon
610 root 20 0 234M 7204 6756 S 0.0 0.4 0:00.00 /usr/libexec/power-profiles-daemon
649 root 20 0 115M 12480 12456 S 0.0 0.6 0:00.09 /usr/bin/python3 /usr/share/unattended-upgrades/unat
676 root 20 0 235M 8508 7832 S 0.0 0.4 0:00.04 /usr/sbin/gdm3
681 root 20 0 235M 8508 7832 S 0.0 0.4 0:00.00 /usr/sbin/gdm3
682 root 20 0 235M 8508 7832 S 0.0 0.4 0:00.00 /usr/sbin/gdm3
699 root 20 0 310M 10448 8844 S 0.0 0.5 0:00.04 gdm-session-worker [pam/gdm-autologin]
EnterSend EscCancel


Activities
Terminal
06:39 26 أبر
yousef@yousef-VirtualBox: ~
yousef@yousef-VirtualBox:~$ df -h
Filesystem Size Used Avail Use% Mounted on
tmpfs 197M 1.6M 196M 1% /run
/dev/sda3 20G 12G 6.8G 63% /
tmpfs 985M 0 985M 0% /dev/shm
tmpfs 5.0M 4.0K 5.0M 1% /run/lock
/dev/sda2 512M 6.1M 506M 2% /boot/efi
tmpfs 197M 176K 197M 1% /run/user/1000
yousef@yousef-VirtualBox:~$ neofetch
LibreOffice Writer
yousef@yousef-VirtualBox
OS: Ubuntu 22.04.5 LTS x86_64
Host: VirtualBox 1.2
Kernel: 6.8.0-110-generic
Uptime: 30 mins
Packages: 1690 (dpkg), 9 (snap)
Shell: bash 5.1.16
Resolution: 1280x800
DE: GNOME 42.9
WM: Mutter
WM Theme: Adwaita
Theme: Yaru [GTK2/3]
Icons: Yaru [GTK2/3]
Terminal: gnome-terminal
CPU: Intel i5-8400H (1) @ 2.496GHz
GPU: 00:02.0 VMware SVGA II Adapter
Memory: 650MiB / 1968MiB
yousef@yousef-VirtualBox:~$

Activities Text Editor 07:29 26 أبر

yousef@yousef-VirtualBox: ~

```
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
yousef@yousef-VirtualBox:~$ python3 -m pip install flask
Defaulting to user installation because normal site-packages is not writeable
Collecting flask
  Downloading flask-3.1.3-py3-none-any.whl (103 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 103.4/103.4 KB 775.6 kB/s eta 0:00:00
Collecting werkzeug>=3.1.0
  Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 226.5/226.5 KB 1.7 MB/s eta 0:00:00
Collecting blinker>=1.9.0
  Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Collecting jinja2>=3.1.2
  Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 134.9/134.9 KB 902.6 kB/s eta 0:00:00
Collecting itsdangerous>=2.2.0
  Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Collecting markupsafe>=2.1.1
  Downloading markupsafe-3.0.3-cp310-cp310-manylinux2014_x86_64.manylinux_2_17_x
86_64.manylinux_2_28_x86_64.whl (20 kB)
Collecting click>=8.1.3
  Downloading click-8.3.3-py3-none-any.whl (110 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 110.5/110.5 KB 2.1 MB/s eta 0:00:00
Installing collected packages: markupsafe, itsdangerous, click, blinker, werkzeu
g, jinja2, flask
```

Open app.py ~/ Save

```
1 from flask import Flask
2 import time, random
3
4 app = Flask(__name__)
5 @app.route('/')
6 def hello():
7
8     delay = random.expovariate(1/0.1)
9     time.sleep(delay)
10    return f"response after {delay:.2f} seconds"
11 app.run(host='0.0.0.0, port=5000')
12
13
```

Home

Activities Text Editor 07:33 26 أبر

Firefox Web Browser yousef@yousef-VirtualBox: ~

```
).
The following packages were automatically installed and are no lo
nger required:
  libpython2-stdlib libpython2.7-minimal libpython2.7-stdlib
  python-pkg-resources python-setuptools python2 python2-minimal
python2.7
  python2.7-minimal
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
yousef@yousef-VirtualBox:~$ python3 -m pip install flask
Defaulting to user installation because normal site-packages is n
ot writeable
Collecting flask
  Downloading flask-3.1.3-py3-none-any.whl (103 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 103.4/103.4 KB 775.6 k
B/s eta 0:00:00
Collecting werkzeug>=3.1.0
  Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 226.5/226.5 KB 1.7 M
B/s eta 0:00:00
Collecting blinker>=1.9.0
  Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Collecting jinja2>=3.1.2
  Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 134.9/134.9 KB 902.6 k
B/s eta 0:00:00
Collecting itsdangerous>=2.2.0
  Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Collecting markupsafe>=2.1.1
  Downloading markupsafe-3.0.3-cp310-cp310-manylinux2014_x86_64.m
anylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (20 kB)
Collecting click>=8.1.3
  Downloading click-8.3.3-py3-none-any.whl (110 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 110.5/110.5 KB 2.1 M
B/s eta 0:00:00
Installing collected packages: markupsafe, itsdangerous, click, b
linker, werkzeug, jinja2, flask
  WARNING: The script flask is installed in '/home/yousef/.local/
bin' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to sup
```

Open plot.py ~/ Save

```
1 import numpy as np
2 import matplotlib.pyploy as plt
3
4 data = np.random.exponential(scale=0.1, size=100)
5
6 plt.hist(data, bins=20, edgecolor='black')
7 plt.title("latency histogram")
8 plt,show()
9
10
```

Python 2 Tab Width: 8 Ln 1, Col 1 INS

Home

Data Center Design Sketch

Internet

|

Load Balancer

|

Core Switches

|

Fat Tree Network

|

Rack 1 Rack 2

| |

Servers Servers

| |

Hypervisor

Docker Runtime

Containers

Cooling + Power Units

Final Write-Up Answers

Q1: Which architecture is better for microservices?

Containers are better because they are lightweight, fast to start, scalable, portable, and consume fewer resources than VMs.

Q2: Why does AWS split Nitro into hardware components?

To improve security, performance, and offload virtualization tasks from CPU to dedicated hardware.

Q3: When use VMs over containers?

Use VMs when stronger isolation, multiple operating systems, or legacy software support is required.

Q4: How does tail latency change with more parallel calls?

As concurrency increases, slow responses become more frequent, increasing tail latency.